SPATHI-benchmarks (FULL)

From parameter search to fair held-out evaluation and scalability testing

Planned workflow for the full synthetic study profile

SPATHI

74 seeded synthetic datasets

554 calibration fits

32 held-out dataset instances

16 routes in 5 comparison families

1 Calibration

Find a robust SPATHI policy on development data

A 7 training scenarios × 2 development replicates

14 synthetic datasets with variation in topology, noise and group balance

Dataset 1 · Dataset 2 · Dataset … · …

- Different topologies
- Varying noise levels
- Different group balance

B 27 candidate parameter sets

Pilot fit with a 50-tree forest

27 parameter sets → 50 trees per fit

378 total fits (screening)

C Advance 4 finalists

Evaluate exact forest sizes: 25, 50, 100, 250 trees

- ✔ Finalist 1
- ✔ Finalist 2
- ✔ Finalist 3
- ✔ Finalist 4

25 · 50 · 100 · 250

Reuse one 250-tree forest with exact prefixes

168 validation fits

8 transfer fits

D Pick one frozen SPATHI policy

Score = 50% moderate validation + 50% transfer robustness

- 2% equivalence band
- Break ties by stability, then smaller forest

Frozen policy

Used for all held-out and scalability experiments

2 Held-out comparison

Evaluate fairly across diverse test datasets

A 30 primary test datasets + 2 deployment-scale

32 held-out dataset instances (unseen in calibration)

30 × Primary test datasets · 2 × Deployment-scale

B Each dataset is sent through every eligible method

Run the frozen policy and comparison methods

Test dataset → SPATHI (frozen policy) · Comparison method 1 · Comparison method 2 · Comparison method … · …

C 5 fair comparison families

Methods grouped by information and assumptions

- Matched information (primary)
- Ordered states
- Lineage-aware
- All-gene fitting
- Topology only

492 eligible route × dataset requests

D Shared metrics and paired statistics

Same metrics for all methods, paired by dataset

- AUPR
- AUROC
- F1 at truth count

Complete paired primary blocks
Friedman + all-pairs Wilcoxon-Holm

3 Scalability

Test computational limits and practical feasibility

A 4 size scales

small → medium → large → extra-large
(4 scales × 3 replicates × 16 routes = 192 measurements)

Small → Medium → Large → Extra-large

192 planned ladder measurements

B Isolated, resource-controlled runs

One method at a time with fixed resources

- CPU (controlled)
- RAM (controlled)

C Thread scaling on the medium dataset

1 CPU, standard CPUs, and all CPUs

Medium dataset → 1 CPU · Standard CPUs (e.g. 4, 8, …) · All CPUs (max)

Reuse the medium baseline when the effective thread count matches.

D Measure runtime, memory, and feasibility frontier

Track performance and stop when infeasible

- Runtime (up to 4 h)
- Memory usage
- Success
- Failed (stop larger sizes)
- Not attempted (blocked runs)

4 h timeout per physical scalability task

If a route fails at one size, stop that route at larger sizes. Later blocked runs are marked not attempted.

Reproducible execution

Semantic seed 734921 · fixed CPU/RAM plan · SQLite journal · atomic artifacts

Portable outputs

HTML reports + CSV tables + provenance files

Synthetic data only · counts are planned protocol quantities, not measured results.